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Article

On the "Staple" Effect in the D-Glucose Acetylation by Choline-Chloride Ethylene Glycol and Choline-Chloride Urea Deep Eutectic Solvents

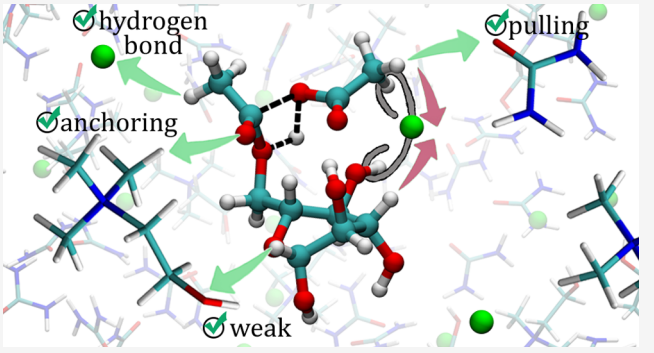
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ABSTRACT: This work employs hybrid quantum mechanics molecular mechanics simulations coupled with an umbrella sampling technique to investigate the acetylation of D-glucose with acetic anhydride in two deep eutectic solvents (DESs): choline-chloride ethylene glycol (ChCl/Etg) and choline-chloride urea (ChCl/U). The reaction proceeds spontaneously, with activation free energy barriers of 18.19 ± 0.15 and 19.83 ± 0.15 kcal/mol for ChCl/Etg and ChCl/U, respectively. These values align with literature data for similar reactions in ionic liquids, highlighting the potential of DESs as green reaction media for wood modifications, while minimizing lignocellulosic matrix degradation. Energy pair distribution analysis, radial distribution functions, and noncovalent interactions reveal a more solvated transition state compared to the initial reactant state within ChCl/Etg, facilitating the acetylation reaction. Interestingly, combined distribution function analysis unveils a "staple" effect in both solvents, potentially hindering efficient product separation. The work further explores hydrogen bond networks and Kamlet−Taft solvatochromic parameters to elucidate the role of solvents in the reaction mechanism. It was observed that a notable decrease in activation energy is accompanied by increasing net basicity, along with a reduction in the "staple" effect. This suggests that solvent parameters could serve as an effective screening tool for identifying novel and efficient DESs for wood modifications.



INTRODUCTION

In the quest for sustainability and a green economy, it is imperative to develop new processes that will aid in the preservation, modification, and elongation of the service life of end products derived from renewable sources.[1] Lignocellulosic biomass (LCB), comprising cellulose, hemicellulose, and lignin, is widely regarded as a key player in the scope of sustainable resources.[2] Wood, a prominent example of such a lignocellulosic material, currently holds a pivotal role in the ongoing shift toward sustainability.[3,4] It is a versatile, easily workable, cost-effective, available, and sustainable material, serving a broad spectrum of purposes ranging from decorative to high-performance applications.[5] Despite the manifold advantages associated with wood and its derivatives, there exist significant challenges. Of notable concern is their inherent hygroscopic behavior, making them highly susceptible to a range of pests and organisms such as fungi, borers, termites, as well as degradation from UV radiation and flammability.[1,5,6] This susceptibility is known to significantly impact the dimensional stability, mechanical properties, and durability of wood, primarily due to its moisture content. Such limitations hinder its full potential as a sustainable material, particularly in the construction sector.[3,7]

Numerous treatments have been proposed to provide new functionalities and enhance the properties of wood.[7] These methods include chemical, mechanical, biological, and physical treatments.[4,6,8] The available wood treatments alter the interaction between water and the structural wood constituents (cellulose, hemicellulose, and lignin), consequently influencing the equilibrium moisture content.[4] Moisture is naturally present in wood due to its hydroxyl and other oxygen-constituent groups, allowing the absorption of water from the environment.[5,7] In this regard, chemical treatments of wood have been extensively studied and are considered one of the most effective methods for improving wood properties.[4−6] Most of the chemical modification treatments of wood involve the replacement of highly hygroscopic hydroxyl groups present within its biopolymer constituents by hydrophobic or bulky reactants through chemical reactions.[5,8] As a result, not only

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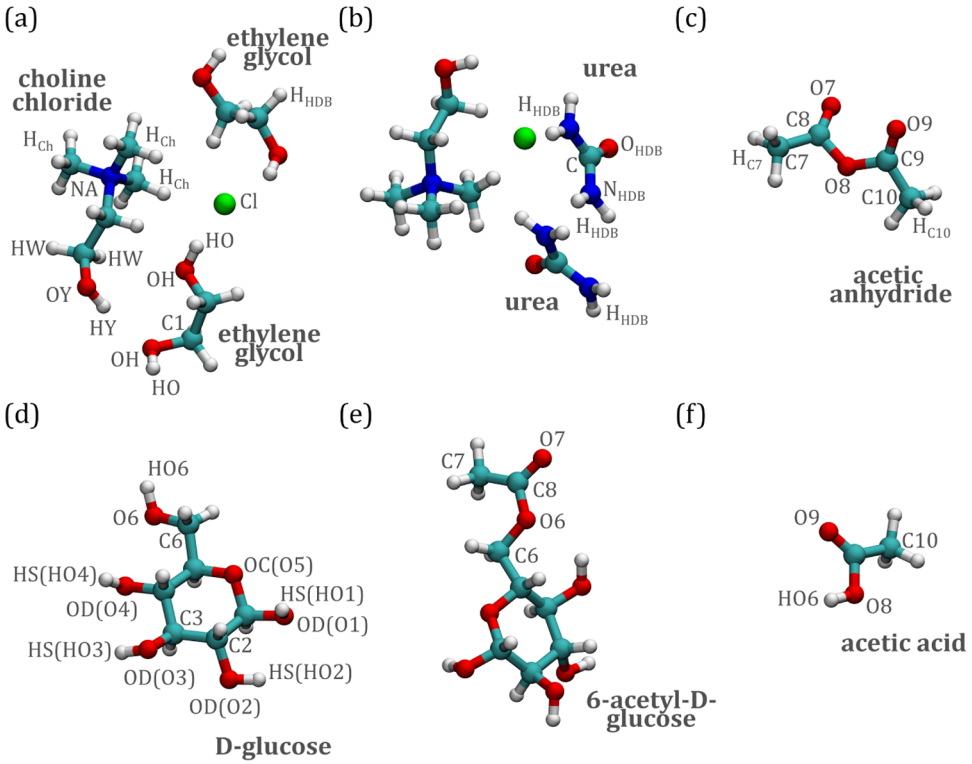


Figure 1. Atomic labels and molecular structures for the deep eutectic solvents, reactants, and products modeled in the present work. (a) Choline-chloride ethylene glycol (ChCl/Etg) at 1:2 HBA/HBD molar ratio, (b) choline-chloride urea (ChCl/U, 1:2), (c) acetic anhydride, (d) β-D-glucopyranose, referred to here as D-glucose, (e) acetylated D-glucose (6-acetyl-D-glucose), and (f) acetic acid. Oxygen atoms in hydroxyl groups from D-glucose, except for O6, are referred to as OD atoms. Similarly, hydrogen atoms in hydroxyl groups (except for HO6) are referred to as HS atoms. Finally, oxygen atoms within the six-membered ring are referred to as OC atoms. Labels in parentheses refer to the typical atom labels used for monosaccharides. The color/atom code is as follows: blue/nitrogen, green/chlorine, gray/carbon, red/oxygen, and white/hydrogen. All images have been created with the visual molecular dynamics (VMD) software.[42]

the wood matrix becomes hydrophobic[6] but also the available space for water molecules within wood is reduced, affecting its equilibrium moisture content.[7]

The acetylation reaction is a well-known and promising method for producing sustainable and high-performance materials from wood.[9,10] Wood acetylation is commonly conducted using acetic anhydride,[1,5,11] due to its broad availability and reactivity.[12] This treatment not only enhances the aforementioned wood properties[4−6,13] but could also potentially contribute to reducing illegal logging and the high demand for tropical hardwood.[3,5,6,9] Furthermore, life cycle analyses show that the acetylated wood has a lower carbon footprint than concrete, steel, and azobe wood, indicating that it is still a sustainable material.[9] In spite of its advantages, the acetylation of wood with acetic anhydride presents certain drawbacks: (i) the use of acetic anhydride can lead to depolymerization of cellulose,[12] lignin degradation, and carbohydrate hydrolysis,[6] which can contribute to the decrease of wood strength;[5] (ii) it produces acetic acid as a byproduct, which can cause equipment corrosion when used at a large scale[8] and further cellulose depolymerization[12] if not removed, and (iii) the rate-limiting step is the degree of penetration of acetic anhydride into the cell wall.[5,14] Therefore, new technologies must be introduced or developed in order to improve the process by reducing the amounts of acetic anhydride, augmenting the capture of acid, and improving the diffusion of acetic anhydride while maintaining the environmentally sustainable aspects of the process.

Deep eutectic solvents (DESs) are a promising class of solvents.[15] DESs consist of a hydrogen bond donor (HBD) combined with a halide or quaternary ammonium salt, referred to as a hydrogen bond acceptor (HBA), at a specific HBA/HBD molar ratio.[15,16] Such a composition results in a eutectic mixture, in which HBA and HBD form hydrogen bonds (HBs), while charge delocalization occurs, allowing the mixture to maintain a liquid state at temperatures below 100 °C.[17] The DESs are regarded as green solvents due to their low vapor pressure, straightforward synthesis, cost-effective raw materials, high biodegradability, and nontoxic properties.[15] DESs are considered as "designer" solvents since tailoring their constituents can lead to the formation of task-specific solvents with unique properties, such as density, viscosity, pH, ionic conductivity, surface tension, refractive index, and hydrogen bonding.[18] Consequently, DESs have found applications in a wide range of fields,[18,19] including the pretreatment and conversion of the LCB into valuable products.[20−23] The capability of DESs to dissolve LCB components is lower compared with that of ionic liquids (ILs).[15] However, this attribute could be advantageous as it may prevent extensive degradation of the highly recalcitrant biopolymer, making DESs suitable as pretreatment solvents for entire wood applications,[24] including the acetylation of simple sugars[22] and cellulose.[25] Unfortunately, there is limited literature on their role in the acetylation of wood or its components,[22,26] in contrast to other solvents such as ILs.[12,27−31]

Since the mechanical properties of wood can be modified through chemical treatments,[3,5,6] and considering that DESs

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have been utilized both as a solvent and catalyst in LCB modification,[22−26] DESs can be proposed as potential media for wood acetylation without degrading additionally wood components. However, the search for an optimal DES that balances yield, selectivity, cost, and minimal wood structural deterioration is time-consuming.[15] Therefore, understanding the intermolecular interactions is crucial to achieving these objectives. In this context, computational modeling, encompassing quantum mechanical (QM) methods, classical molecular dynamics (MD) simulations, and machine learning (ML) algorithms, has proven to be a powerful tool for gaining fundamental knowledge about the atomistic interactions that govern macroscopic behavior of materials.[16] Despite the limited information regarding the role of DESs and ILs as both catalyst and solvents in the acetylation of LCB from an atomistic perspective, several authors[14,30,32,33] have endeavored to address this issue. This research aims to address this gap by investigating the role of DESs as a medium in the acetylation reaction, building upon the insights gained from our previous work.[34] Molecular simulations can provide an atomistic understanding of not only the interactions that govern and influence the macroscopic behavior of DESs as pure substances[16] but also their role as reaction media and catalysts.[18] However, calculations involving explicit solvent molecules have received less attention due to the computational demands associated with sampling both the solute and solvent degrees of freedom.[35] To the best of our knowledge, a specific study utilizing a hybrid quantum mechanics/molecular mechanics (QM/MM) approach to investigate the impact of choline-chloride (ChCl)-based DESs in the context of the acetylation reaction has yet to be conducted.

This study employs hybrid QM/MM calculations to elucidate the role of ChCl/ethylene glycol (ChCl/Etg, 1:2) and ChCl/urea (ChCl/U, 1:2) DESs (Figure 1a,b) in the acetylation of β-D-glucopyranose (D-glucose), a model compound for cellulose, with acetic anhydride (Figure 1c). Choline chloride (ChCl) was selected as the HBA owing to its economic viability, biodegradability, and low toxicity profile.[23,36] ChCl/Etg and the ChCl/U DESs were chosen as reaction media due to their sustainable characteristics.[21] These DESs have been shown to promote cellulose dissolution while sparing LCB matrix,[34,37,38] thereby preserving the wood strength. The D-glucose molecule, a monomer of cellulose (Figure 1d), was selected as a suitable model compound for cellulose because the hygroscopic behavior of wood is predominantly localized within the cell walls,[7] where cellulose is the main constituent.[39] Furthermore, structural analysis reveals a noteworthy "staple" effect in both solvents, which could potentially hinder efficient product separation.

THEORETICAL METHODS

Force Fields and Level of Theory. ChCl/Etg (1:2) and ChCl/U (1:2) DESs were modeled using the Optimized Potentials for Liquid Simulations force field (FF) for deep eutectic solvents (OPLS-DES), developed by Doherty and Acevedo.[40] While the OPLS-DES FF was originally developed for the GROMACS software suite, all simulations in this work were conducted using the AMBER package due to its comprehensive tools and built-in modules for QM/MM simulations. This eliminates the need for an interface with external quantum chemistry software. Consequently, the OPLS-DES FF was adapted to ensure compatibility with AMBER.[41] Specifically, for the ethylene glycol HBD, Doherty and Acevedo included intramolecular exclusions between nonbonded HO and OH atoms (refer to Figure 1a for labels). Although exclusions are supported within the AMBER package, the MD engine of AMBER, sander, overwrites the explicit exclusions in the parameter-topology file. To assess the effect of omitting such exclusions on the calculated properties of ChCl/Etg DES in AMBER, density, enthalpy of vaporization, and radial distribution functions (RDFs) were computed and compared with those obtained from GROMACS simulations (details are provided in the Supporting Information). The analysis revealed that omitting intramolecular exclusions between OH and HO atoms in the ethylene glycol DES is unlikely to substantially impact the results of subsequent calculations.

In the hybrid QM/MM MD simulations, the MM region comprised the DESs, whereas the QM section included the D-glucose and acetic anhydride (reactants, Figure 1c,d) and the 6-acetyl-D-glucose and acetic acid (products, Figure 1e,f). The QM region was modeled with the semiempirical (SE) level of theory parametric method number 6 with energy dispersion and HB corrections (PM6-DH+).[43−46] The SE level of theory was chosen over the DFT level due to the extensive phase-space sampling required to ensure accurate computation of the free energy differences.[47,48] This SE level of theory has been extensively employed in the literature to model chemical reactions in both condensed and gas phases,[49−52] demonstrating its efficacy in modeling organic compounds, including cellulose[53] and monosaccharides,[54] such as glucose.[55−57]

To complete the simulation setup, the Lennard-Jones (LJ) parameters (ϵ_i and σ_i) for D-glucose and acetic anhydride were taken from the OPLS 2005[58] FF to ensure consistency with those from the OPLS-DES FF. These LJ parameters were assigned using the Maestro[59] interface within the Desmond[60] MD program.

QM/MM Simulation Details. All simulations were performed using the AMBER simulation package.[41] The QM/MM interface developed by Walker et al.[61] was used for both QM and hybrid QM/MM MD simulations. The Langevin Dynamics[62] thermostat was employed to maintain a constant temperature of 100 °C (373.15 K). This temperature value was selected because a number of acetylation reactions in both ILs and DESs have been reported around 100 °C.[5,6,26,29] Moreover, conducting the acetylation reaction at lower temperatures is generally time-consuming, particularly on a large scale.[6] The pressure, when necessary, was maintained at 1 bar using the Berendsen[63] barostat. The Berendsen barostat has been extensively employed to compute both thermodynamic and kinetic properties, consistently yielding results in agreement with the experimental data.[40,49] The collision frequency for temperature and pressure relaxation time were 5 ps^{-1} and 2 ps, respectively. Newton's equations of motion were integrated using the Leapfrog integrator with a time step (Δt) of 1 fs. Periodic boundary conditions (PBC) with a cutoff value of 12 Å (for both QM and MM regions) were implemented to calculate nonbonded interactions for all condensed-phase simulations. For the gas-phase simulations, nonperiodic conditions with an infinity cutoff were used. Electrostatic interactions beyond the cutoff were computed using the particle-mesh-Ewald (PME) algorithm[64] for condensed-phase simulations, while van der Waals (vdW) interactions were truncated at the cutoff for both condensed- and gas-phase simulations. The SHAKE[65] algorithm was utilized to constraint all covalent bonds involving hydrogen

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atoms in the DESs (MM region). It is important to note that the SHAKE algorithm was not applied to the QM region. The LJ mixed interactions (ϵ_{ij} and σ_{ij}) were computed using geometric combining rules. The intramolecular 1−4 LJ and 1−4 electrostatic interactions were scaled by a factor of 0.5.

Finally, the interactions between QM and MM regions were exclusively handled through electrostatic and vdW interactions, as they are not covalently bonded. Electrostatic interactions between regions were addressed through electronic embedding, enabling polarization of the QM electron density in response to the electrostatic MM environment. Conversely, the vdW interactions were computed using the classical LJ potential.[66]

Minimum Energy Path. The minimum energy path (MEP) represents the most probable transition path between reactants and products, which correspond to local minima on the *n*-dimensional potential energy surface (PES).[67−69] Identifying the MEP defines the reaction coordinate (ξ), which can be a geometrical, energetic, or other measurable parameter describing the transformation from reactants to products.[70] This allows the calculation of the one-dimensional free energy change, known as potential of mean force (PMF).[70,71] For this study, energy minimizations were performed on the reactant (D-glucose and acetic anhydride) and product (acetylated D-glucose at the C6 position and acetic acid) structures in various conformations. The C6 position in D-glucose (Figure 1d) was selected due to its susceptibility to acetic anhydride attack, as reported in several studies.[30,72−75] HBDs facilitate proton dissociation and complex formation with the reactants. However, experimental studies have shown that ethylene glycol and urea HBDs do not provide a sufficient concentration of H^+ to significantly alter the neutral mechanism in other reactions.[76] Therefore, HBA and HBD were excluded from the MEP calculation in this study.

The minimization of the systems was performed at the PM6-DH+ SE level of theory for 10,000 steps using the *steepest-descent* algorithm. The conformation with the lowest potential energy for both the reactant and product (Figure S3) was selected in the MEP calculation employing the partial nudged elastic band (p-NEB) algorithm.[77] The obtained MEP is depicted in Figure S4 (details are provided in the Supporting Information). From that figure, the entire reaction can be described by three distances: d_{C8-O8}, d_{C8-O6}, and d_{C8-HO6}. However, a one-dimensional (1D) free energy calculation is preferred. Combining the distances, the generalized reaction coordinate (ξ) is proposed

$$\xi = d_{C8-O8} - d_{C8-O6} \quad (1)$$

In this expression, the d_{C8-O8} distance from acetic anhydride controls the C8−O8 bond breaking, while the distance between the C8 carbon atom and the O6 oxygen atom from D-glucose (d_{C8-O6}) governs the approach of the two molecules, ensuring the formation of the C8−O6 bond and the capture of the HO6 hydrogen by the O8 oxygen atom.

QM/MM MD Simulations. Boereboom et al.[52] emphasize the importance of representing the solvent with explicit molecules to capture the steric environment and specific atom−atom interactions that occur between the MM and QM regions. In this context, the reactant and product in the condensed phase were constructed using the Packmol[78] package as follows: The center-of-mass (COM) of the minimized reactant (D-glucose and acetic anhydride, Figure S3a) and the COM of the minimized product (acetylated D-glucose and acetic acid, Figure S3b) were positioned at coordinates x = 6 Å, y = 6 Å, and z = 6 Å within a cubic box with a length size of 42 Å. Subsequently, 157 molecules of ChCl and 314 molecules of HBD (ethylene glycol or urea) were placed within the simulation box. The QM region, modeled at the PM6-DH+ SE level of theory, included the reactant and product, while the MM region, modeled with the OPLS-DES FF, consisted of the DES molecules. This hybrid theoretical level is also referred to as the PM6-DH+/OPLS-DES. The QM/MM protocol employed in this study, which involves selecting only the chemically active or reactant part as the QM region and treating the solvent as the MM region, has been widely used by numerous authors to calculate free energy differences.[49,79−84]

The objective of the QM/MM MD simulations was to obtain the initial and final values of the reaction coordinate (eq 1) for the condensed phase. To achieve this, the reactant-DES and product-DES systems were minimized using 10,000 steps of the *steepest-descent* algorithm to eliminate any steric clashes or bad contacts. Subsequently, QM/MM MD simulations were carried out for 2 ns in the NPT ensemble to equilibrate the systems. To prevent simulation instabilities resulting from the expansion of the QM region due to reactant or product diffusion, the coordinates of the C5 atom from D-glucose and C10 from acetic anhydride were restrained in the reactant-DES systems. The C10 atom from acetic anhydride was chosen because its restraint allows the free movement of the active atoms involved in the reaction. Similarly, the C5 atom coordinates were restrained in the product-DES systems. However, in this case, the C10 carbon atom from acetic acid was restrained at 7 Å from the C5 carbon atom of acetylated D-glucose. The restraint was implemented by using a harmonic biasing potential with a force constant of 20 kcal/mol/Å^2. After NPT equilibration, a QM/MM MD simulation in the NVT ensemble was performed for both the reactant-DES and product-DES systems for 2 ns. The restraint imposed on the C5−C10 distance during the NPT simulation was removed in the NVT simulation. The energy convergence over time along the NVT simulation is shown in Figure S5. The equilibrated conformations of the reactant-DES and product-DES systems after NPT−NVT simulations at 373.15 K are displayed in Figure S6.

QM and QM/MM MD Umbrella Sampling Simulations. The PMF calculation for the acetylation of D-glucose in DESs was computed with the umbrella sampling (US) method.[47,70] Briefly, in the US method, the reaction coordinate is discretized into a series of reference values (ξ_i^{ref}) between the initial (reactant) and final (product) values. Each ξ_i^{ref} represents an US calculation of the window i (W_i) in which ξ is restrained and biased to the ξ_i^{ref} value by means of a harmonic bias function, $w_i(\xi)$, defined as[70]

$$w_i(\xi) = \frac{k}{2}(\xi - \xi_i^{ref})^2 \quad (2)$$

Here, k represents the bias force constant. To obtain a reliable PMF calculation through US methodology, the phase space of the system around ξ_i^{ref} must be properly sampled. For this reason, the US method is usually combined with the MD simulation at the SE theoretical level to ensure exhaustive sampling.[47,70,85]

US MD simulations were conducted for both gas- and condensed-phase systems. In the gas-phase system, the

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reaction coordinate ($d_{C8-O8}-d_{C8-O6}$) ranged from −2.20 Å (reactant, Figure S4a) to 3.02 Å (product). The ξ value of the product was extended beyond the obtained value from the minimized product shown in Figure S4f (2.23 Å) to observe changes in the calculated PMF at larger distances. Kästner and Thiel[86] determined that the error in the free energy calculations increases if the distances between consecutive windows ($\Delta\xi$) are larger than $3/\sqrt{\beta k}$, with $\beta = 1/k_BT$. For a temperature value of 373.15 K and k = 750 kcal/mol/Å^2, $\Delta\xi$ is approximately 0.09 Å. However, previous calculations[49] found that sampling the regions close to the transition state requires either a larger number of US windows or larger values of k. In this regard, a value of 0.06 Å for $\Delta\xi$ was used along with k = 750 kcal/mol/Å^2, providing a total of 88 US windows between −2.20 Å and 3.02 Å for the gas-phase reaction. Each US window had a length of 250 ps, with ξ values saved every 5 fs.

To enhance ξ convergence[66] and the appropriate approximation of acetic anhydride to the D-glucose molecule, the coordinates of the C5 carbon atom from D-glucose were restrained. Additionally, the following procedure was implemented: First, the O6−HO6−O8 angle was restrained at 110° by using angular harmonic restrictions with a bias force constant of 50 kcal/mol/rad^2. Second, the output coordinates from the first US window were used as input coordinates for the second US window, then the third US window used the output coordinates from the second US window and so forth, creating a series of sequential simulations.

In the condensed-phase systems, reaction coordinate values for the reactant and product were obtained from the QM/MM MD simulations (Figure S6). Accordingly, the ξ ranged from −2.26 Å (reactant) to 3.50 Å (product). A total of 97 US windows were required to sample the acetylation of D-glucose in DESs using $\Delta\xi$ = 0.06 Å and k = 750 kcal/mol/Å^2. Similar restraints to those applied in the gas-phase US MD simulations were used. The equilibrated coordinates of the reactant-DESs from the QM/MM MD NVT simulations were used as input coordinates for the first QM/MM-US window. In the first stage, each US window was equilibrated for 50 ps in the NVT ensemble in a sequential manner, saving the ξ values every 5 fs. Once the entire reaction coordinate was simulated, a second run of 450 ps length was carried out for each US window using the output coordinates of the corresponding 50 ps US window, resulting in a total simulation time of approximately 50 ns for the entire trajectory. Since all of the input coordinates for the second run were available, this series of QM/MM-US MD simulations was performed in parallel. The QM/MM structural and energetic analyses were conducted using data from the final 150 ps of the respective reactant and transition-state US windows.

Potential of Mean Force. The probability distribution of the reaction coordinate around ξ_i^{ref} was obtained by analyzing the trajectory data from the last 125 ps of the QM-US MD simulations in the gas phase and the last 150 ps of the QM/MM-US MD simulations in the condensed phase. The integration of these histograms provides the PMF of the reaction. In this work, the PMF was obtained using the weighted histogram analysis method (WHAM).[87] Since the accurate integration of the PMF through WHAM analysis requires the overlap of the histograms along the reaction coordinate,[70] the overlap was corroborated before performing the corresponding PMF integration. Subsequently, the PMF for gas-phase and condensed-phase reactions were obtained using the WHAM script developed by Grossfield,[88] employing 200 bins between the initial and final values of the reaction coordinate. The integrations were evaluated with a tolerance of 1×10^{-6} kcal/mol, and statistical uncertainties on the PMF were determined through the bootstrap method.[89]

RESULTS AND DISCUSSION

Gas-Phase Reaction. Selected configurations from the MEP are depicted in Figure S4. As observed from Figure S4a,b, the distance between the O6 oxygen atom from D-glucose and the C8 carbon atom from acetic anhydride consistently decreases across the images. This interaction weakens the C8−O8 bond, which consequently increases its length, as shown in Figure S4c. Simultaneously, as the C8−O6 distance decreases, the O6−HO6 distance increases. These concurrent effects facilitate the interaction between the HO6 hydrogen and the O8 oxygen atoms, resulting in the formation of the corresponding bond, as illustrated in Figure S4d,f. This mechanism, following nucleophilic substitution, induces the displacement of the HO6 atom from the alcohol moiety toward the acetate moiety of acetic anhydride. A similar mechanism has been reported by several authors.[90−93]

The US histograms, representing the reaction coordinate probability distribution ($P(\xi)$), illustrate the overlap in the gas-phase reaction obtained from QM-US MD simulations, as shown in Figure 2. This figure reveals a smooth transition of

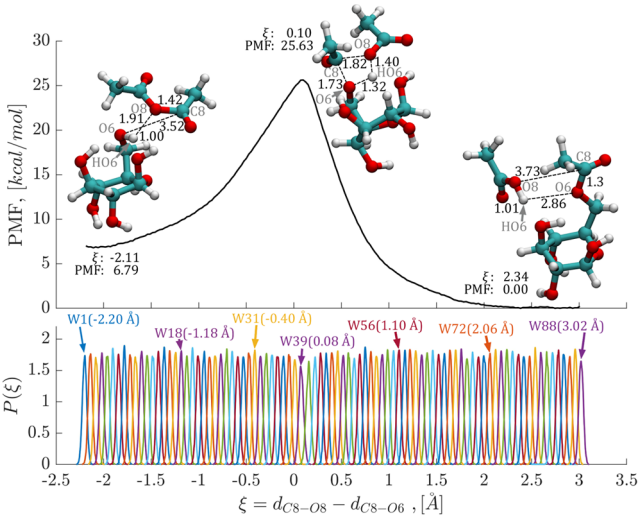


Figure 2. Calculated PMF from the QM-US simulations for the acetylation of D-glucose with acetic anhydride in the gas phase at 373.15 K. The Gibbs free energy profile along the reaction coordinate is displayed in the top panel along with the configurations of the reactant, transition state, and product. Labels of the main atoms involved in the reaction are also provided. The reaction coordinate histogram distribution at ξ_i^{ref} is shown in the bottom panel. The corresponding ξ_i^{ref} values for the selected US windows are displayed in parentheses.

histograms along the reaction coordinate, indicating that the conversion of D-glucose into acetylated D-glucose can be characterized as a continuous process. The histogram plot reveals a slight decay of the $P(\xi)$ around $\xi \sim 0.08$ Å (W39). This effect arises due to the system attaining a specific configuration that is difficult to sample. Since the same force constant value was employed for all US windows in this

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transformation, this behavior could suggest the location of the transition state.

After the overlap for all histograms was confirmed, the PMF was obtained by integrating $P(\xi)$ along the reaction coordinate, as shown in Figure 2. The corresponding activation free energy ($\Delta G^{\ddagger}$) and reaction free energy (ΔG_{rxn}) barriers are summarized in Table 1. The reactant, transition state, and

Table 1. Calculated Activation Free Energy ($\Delta G^{\ddagger}$) and Reaction Free Energy (ΔG_{rxn}) Barriers (kcal/mol) for the Acetylation Reaction of D-Glucose Evaluated in This Work[a]

solvent	$\Delta G^{\ddagger}$	ΔG_{rxn}
gas phase	18.84 ± 0.13	−6.79 ± 0.10
ChCl/Etg	18.19 ± 0.15	−8.00 ± 0.12
ChCl/U	19.83 ± 0.15	−9.24 ± 0.12

[a]At 373.15 K and PM6-DH+ and PM6-DH+/OPLS-DES level of theory. Calculated barriers include the uncertainty represented by one standard deviation.

product configurations, extracted from US MD simulations, are presented in Figure 2 along with their respective ξ and PMF values. Notably, the transition state is identified at ξ = 0.098 Å, within the W39 histogram, consistent with the $P(\xi)$ decay observed from the histogram analysis.

The ΔG_{rxn} barrier from Table 1 exhibits a negative value of −6.79 ± 0.10 kcal/mol, indicating the spontaneity of the acetylation reaction, consistent with previous studies.[14,90,94] Ponnuchamy et al.[14] evaluated the ΔG_{rxn} barrier at 298.15 K for the acetylation of cellobiose (D-glucose dimer) at different carbon positions (C2, C3, and C6 in Figure 1d) using the wB97X-D[95]/6-311G(d,p) level of theory. They reported a value of −19.26 kcal/mol for acetylation at the C6 position. Due to the exothermic nature of this reaction, it is expected that ΔG_{rxn} increases (becomes more positive) when the temperature increases. This behavior is consistent with the calculated ΔG_{rxn} value in the present work. To scrutinize the reliability of the selected SE level of theory employed in the current study, the ΔG_{rxn} was also computed at the gas phase using the M06[96]/6-311G(d,p) level of theory in the Gaussian16[97] software. The ΔG_{rxn} was found to be −3.84 kcal/mol at 373.15 K, in qualitative agreement with the value obtained at the PM6-DH+ SE level of theory. Interestingly, from Figure 2, it can be observed that the ΔG_{rxn} at larger ξ values did not change significantly in the gas phase. This is because the PMF value at ∼3 Å (0 kcal/mol) is virtually identical to that obtained at ∼2.3 Å (0.14 kcal/mol), as illustrated in Figure 2.

The activation free energy barrier ($\Delta G^{\ddagger}$) obtained in this work is 18.84 ± 0.13 kcal/mol. This value is consistent with the noncatalyzed acetylation reaction of LCB reported by Hill et al.[93] as 12.17 ± 0.73 kcal/mol. Interestingly, utilizing their kinetic initial rate data around 100 °C (373.15 K), the estimated activation energy was 17.46 kcal/mol, in good agreement with the calculated value in the present work. Regarding theoretical calculations, Beaumont et al.[98] calculated the Gibbs free energy barriers for the acetylation of methanol with *N*-acetylimidazole in both aqueous and anhydrous environments. Their computations were conducted at the BP86[99,100]/def2-SVP level of theory with Grimme's D3[101] dispersion corrections. The authors reported an activation barrier of 27.7 kcal/mol, which is approximately 9 kcal/mol higher than the value reported in this work.

Condensed-Phase Reaction. Equilibrated configurations at 373.15 K, following NVT QM/MM MD simulations for both reactant and product within ChCl/Etg and ChCl/U DESs are shown in Figure S6. The obtained histogram overlaps for $P(\xi)$ corresponding to the acetylation reaction of D-glucose in ChCl/Etg and ChCl/U DESs are depicted in Figures 3 and

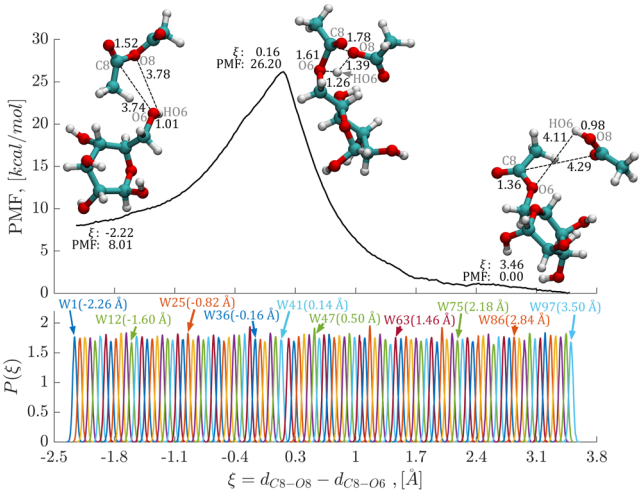


Figure 3. Calculated PMF from the QM/MM-US simulations for the acetylation of D-glucose with acetic anhydride in ChCl/Etg DES at 373.15 K. The Gibbs free energy profile along the reaction coordinate is displayed in the top panel along with the configurations of the reactant, transition state, and product. To enhance clarity, the DES molecules have been omitted, and labels of the main atoms involved in the reaction are provided. The reaction coordinate histogram distribution at ξ_i^{ref} is shown in the bottom panel. The corresponding ξ_i^{ref} value for selected US windows is displayed in parentheses.

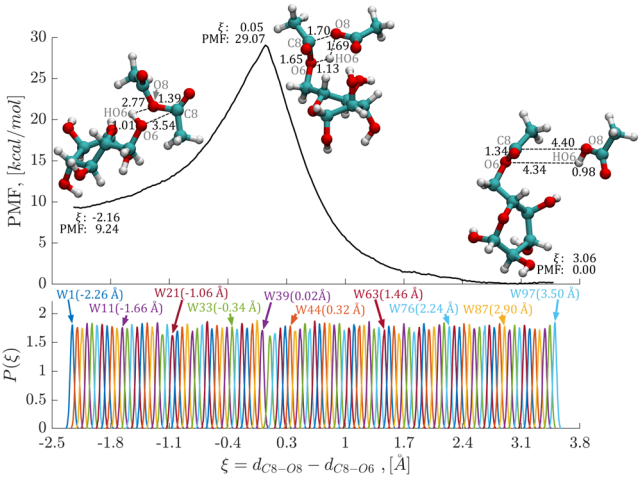


Figure 4. Calculated PMF from the QM/MM-US simulations for the acetylation of D-glucose with acetic anhydride in ChCl/U DES at 373.15 K. Same description as in Figure 3.

4. The conversion of reactants into products is observed as a continuous process, as evidenced by the degree of overlap along the reaction coordinate. Interestingly, the expected decay of the $P(\xi)$ around $\xi \sim 0.16$ Å (Figure 3) is less pronounced for the ChCl/Etg DES than anticipated from the gas-phase calculations (Figure 2). Nevertheless, a slight histogram

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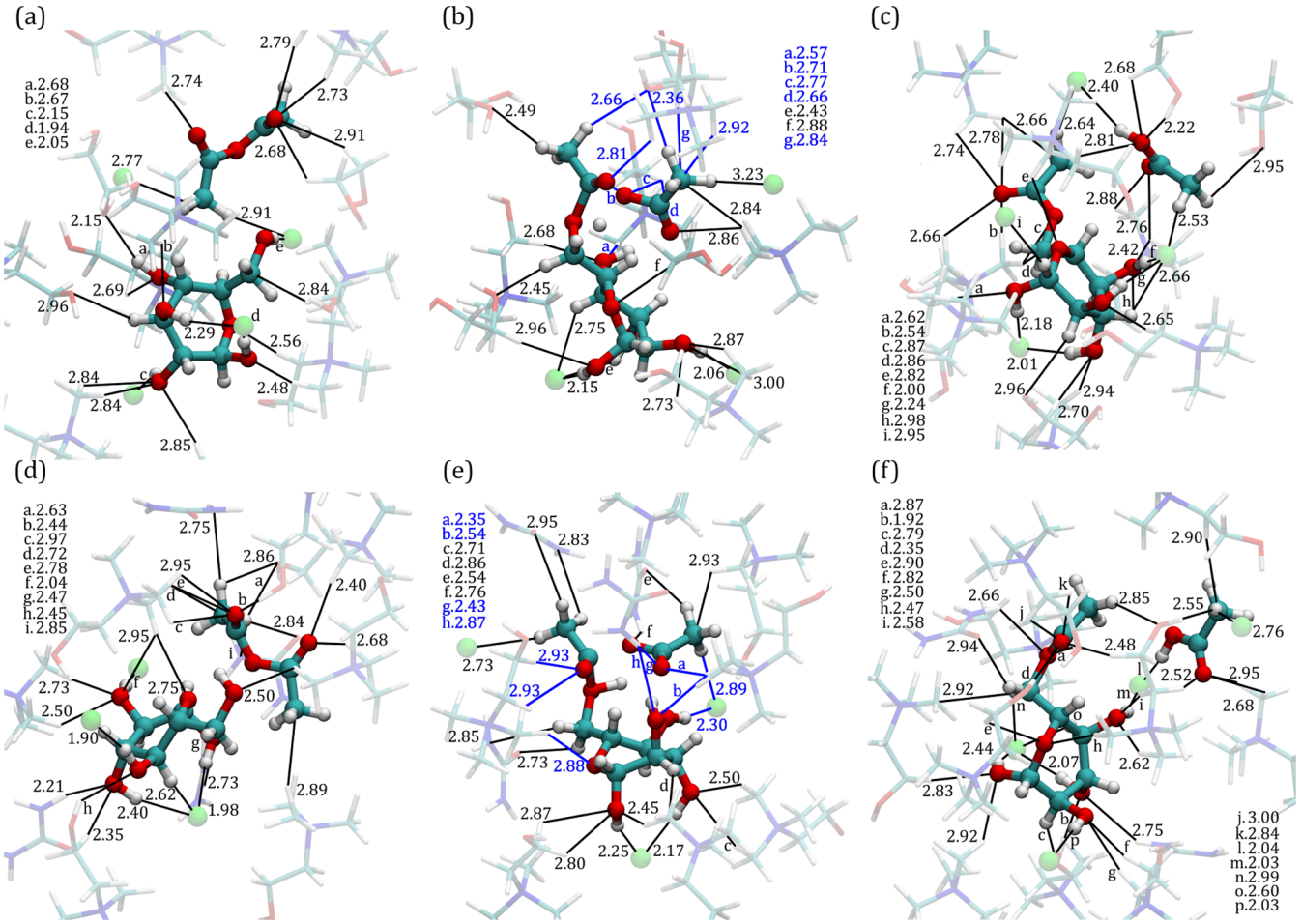


Figure 5. Thermodynamic states for D-glucose acetylation with acetic anhydride in DES: (a) reactant, (b) transition state, and (c) product in ChCl/Etg DES; and (d) reactant, (e) transition state, and (f) product in ChCl/U DES, at 373.15 K. All of the configurations were extracted from the QM/MM-US MD simulations. The closest DES molecules are also included. The QM region is represented in the ball-and-stick style, while the MM region is depicted in licorice representation. Interactions involved in the "staple" effect are highlighted in blue (see text for details). Only relevant distances (Å) between regions are indicated. The color/atom code is as follows: blue/nitrogen, green/chlorine, cyan/carbon, red/oxygen, and white/hydrogen. All images were generated using the VMD software.[42]

separation is evident between simulation windows 41 (W41) and W42. This suggests that the region around $\xi \sim 0.16$ Å (likely corresponding to the transition state) is efficiently sampled with the chosen force constant (k). Conversely, a significant decrease in $P(\xi)$ is observed around $\xi \sim 0.05$ Å for the ChCl/U DES (Figure 4). Considering that the same value for the biased force constant was used for all of the QM/MM-US MD simulations, it can be suggested that the sampling of the reaction coordinate related to the transition state in the ChCl/Etg DES is more extensive than in the ChCl/U DES.

The PMF profiles for the conversion of D-glucose into acetylated D-glucose in DESs are shown in the top panels of Figures 3 and 4, while the calculated free energy barriers in both solvents are listed in Table 1. Notably, the larger decay in the $P(\xi)$ around $\xi \sim 0.05$ Å for the ChCl/U DES correlates with the higher activation free energy barrier (19.83 kcal/mol) compared with the value obtained for the ChCl/Etg DES (18.19 kcal/mol). The observed higher barrier in the ChCl/U DES aligns with the low yields reported by Abbott et al. for the acetylation of monosaccharides.[22] Nawaz et al.[30,32] evaluated the activation parameters, including the $\Delta G^{\ddagger}$, for the acetylation of microcrystalline cellulose (MCC) in a series of IL/molecular solvent mixtures. The reported activation barriers by the authors ranged from 23.1 to 25.82 kcal/mol for both mixtures at 60 °C. Ramsden and Blake[102] reported the acetylation rates of cellulose, hemicellulose, and lignin components in various wood species using acetic anhydride in xylene as the solvent at 373 K. They reported activation energy barriers ranging from 10.04 ± 1.91 kcal/mol (for southern yellow pine) to 25.57 ± 0.48 kcal/mol (for larch). Even though the studies mentioned above employed different and efficient solvents, their reported activation energy barriers are within the same order of magnitude as the calculated $\Delta G^{\ddagger}$ values in the present work. While there is a lack of investigation at the QM/MM level of theory for this study, several theoretical approaches from the literature[33,98,103] allow for comparisons with the performance of the PM6-DH+/OPLS-DES FF level of theory.

Vicente et al.[33] reported the free energy barriers for the deacetylation reaction of chitin within a series of ChCl-based DESs at the M06-2X[96]/6-311++G(d,p) level of theory. Employing the universal continuum solvation model SMD[104] to represent the dielectric medium of the solvent, the researchers found activation barriers ranging from 19 to 40 kcal/mol for both concerted and two-step mechanisms. Interestingly, the values for the concerted mechanism align

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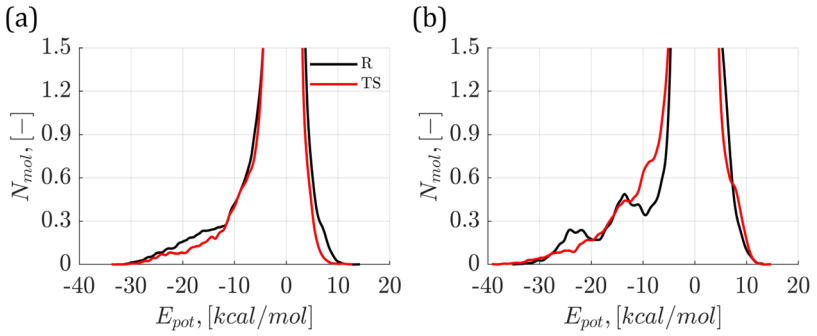


Figure 6. Energy pair distribution (EPD) for reactant (R) and transition (TS) states for the acetylation reaction of D-glucose in (a) ChCl/Etg and (b) ChCl/U DESs.

with the activation barriers observed in the reverse direction in the current study of ∼27.64 kcal/mol. In another relevant study, Beaumont et al.[98] calculated the $\Delta G^{\ddagger}$ of methanol (as a model compound for cellulose) and *N*-acetylimidazole in the presence of water molecules (as solvent) at the BP86-D3/def2-SVP level of theory, reporting a barrier of 15.5 kcal/mol. Furthermore, Oberlintner et al.[103] investigated the acetylation of cellotriose (a D-glucose trimer) with acetic anhydride at M06-2*X*/6-31+G(d)//M06-2*X*/6-311++G(d,p) level of theory using the SMD approach. By employing pyridine as a catalyst with a dielectric constant value of ∼13, similar to those theoretically reported for ChCl/U DES,[105] the authors identified two feasible mechanisms. The mechanism involving no proton transfer resulted in an activation barrier of around 23.9 kcal/mol, consistent with the activation barrier computed in the present work.

While the aforementioned calculations provide a basis for comparing the reliability of QM/MM simulations with the PM6-DH+ SE level of theory, it is important to acknowledge that SE methods typically yield an accuracy of greater than 5 kcal/mol in various thermochemical applications, as noted by Bursch et al.[106] Given this potential variation, the computed free energy barriers at the SE level of theory are still considered promising, as indicated by Vicente et al.[33]

In accordance with the literature, the reaction mechanism follows a nucleophilic pathway, leading to a decrease in the degrees of freedom of the transition state (TS) compared with the reactant state (R). This results in a negative value for the entropic contribution $T\Delta S^{\ddagger}$ to the activation energy, $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$, where $\Delta S^{\ddagger} = S_{TS} - S_{R}$.[107] On the other hand, favorable activation enthalpy values ($\Delta H^{\ddagger} < 0$ with $\Delta H^{\ddagger} = H_{TS} - H_{R}$) indicate that H_{TS} should be more negative (more stable) than H_{R}. As a result, a well-solvated transition state compared to a less-solvated reactant state will decrease the value of $\Delta G^{\ddagger}$.[108] In this context, a lower activation free energy $\Delta G^{\ddagger}$ can be achieved when a favorable $\Delta H^{\ddagger}$ compensates for the $-T\Delta S^{\ddagger}$ term. To gain insights into this observation, the configurations of the reactant, transition state, and product as well as the closest DES molecules surrounding the QM region were extracted from the corresponding QM/MM-US windows and displayed in Figure 5. The transition state was further identified through vibrational frequency analysis conducted at PM6-DH+ SE level of theory in the MOPAC package version 22.0.6.[109]

In Figure 5a,d, the reactant state in the ChCl/Etg DES exhibits less solvation compared to that in the ChCl/U DES. This trend is also observed for the acetic anhydride molecule. Figure 5b,e reveals a similar number of interactions established by molecules from both DESs with the transition state. Consequently, the acetylation reaction is expected to proceed at a slower rate in the ChCl/U DES as the reactant state requires more energy to disrupt its interactions and to reach the transition state.

Upon reaching the transition state, product formation becomes an enthalpy-favorable process. This is attributed to the increased number of interactions between the QM region and the DES molecules in both solvents at the product state, as evidenced in Figure 5c,f. Notably, the greater solvation of the product in ChCl/U DES likely contributes to the lower ΔG_{rxn} value reported in Table 1, compared with the ChCl/Etg DES.

To gain insights into the energetics associated with reactant and transition states, the energy pair distribution (EPD) between the QM and MM regions was computed in both solvents. Briefly, the EPD provides information regarding the average number of molecules interacting with a specific particle at a given potential energy value.[110] Accordingly, the nonbonded energy between these regions was calculated through the LJ and Coulombic potentials as applied in classical FF. While the atomic charges of the MM region are readily provided by the OPLS-DES force field, the Mulliken atomic charges of the QM region were calculated throughout the QM/MM-US trajectory. This was achieved by using the *printcharges* option in the AMBER configuration input file. The computed reactant- and transition-state EPDs are displayed in Figure 6 for both solvents.

As illustrated in Figure 6, a limited number of solvent molecules exhibit strong attractive interactions with both the reactant (D-glucose and acetic anhydride) and transition states within both ChCl/Etg and ChCl/U DESs. In the case of the reactant and transition states in ChCl/Etg DES (Figure 6a), similar strong attractive interactions are observed within the range of −22.5 to −32.5 kcal/mol. Moreover, the stronger interactions of solvent molecules with the reactant state, ranging from −12 to −22.5 kcal/mol, may contribute to the relatively large activation barrier observed in the present study. In contrast to the transition state, the EPD for the reactant in ChCl/U DES (Figure 6b) exhibits two distinct peaks, representing attractive interactions at energy levels of −13 and −24 kcal/mol. Furthermore, while the transition state shows interactions as strong as −39 kcal/mol and displays more substantial interactions around −9 kcal/mol, the distinct and well-defined interactions observed in the reactant state hinder the formation of the transition state.

Figure 5 illustrates differences in interactions involving D-glucose and acetic anhydride between the reactant states (Figure 5a,d) and the transition states (Figure 5b,e). To

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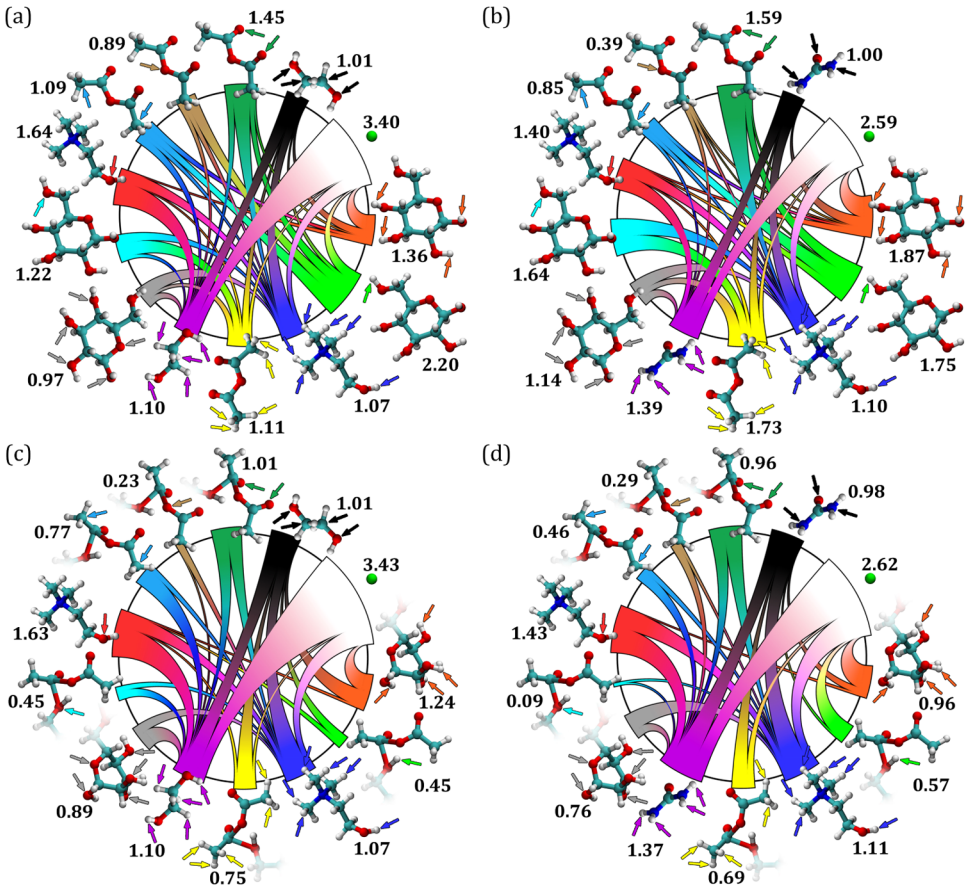


Figure 7. Sankey diagram for average HB count (in bold) per donor−acceptor in (a) reactant-ChCl/Etg DES, (b) reactant-ChCl/U DES, (c) transition-state-ChCl/Etg DES, and (d) transition-state-ChCl/U DES systems. The HBs were computed using a general cutoff distance of 3.5 Å.[117] Arrows indicate the atoms involved in the formation of HBs. The width of the connections between molecules is proportional to the HB count for all donor−acceptor pairs in each diagram.

further investigate these differences, EPDs analysis was performed for both D-glucose and acetic anhydride moieties in the reactant and transition states for each solvent system (Figure S7). As observed, solute−solvent interactions in the reactant state were weaker in ChCl/Etg DES compared to those in ChCl/U DES (Figure S7a,b). Upon transition-state formation, interactions between the acetic anhydride moiety and ChCl/Etg DES intensified, ranging from −21.7 to −34.2 kcal/mol, in contrast to weaker interactions observed in ChCl/U DES (Figure S7c). Conversely, for the D-glucose moiety, a prominent peak was observed at −30.7 kcal/mol in the ChCl/U DES, with some interactions reaching −60 kcal/mol (Figure S7d). Considering these trends, it appears that additional effects are influencing the transition state, potentially either facilitating or impeding specific interactions within each DES.

A closer examination of Figure 5b,e reveals that certain QM-MM interactions involve atoms from both the acetic anhydride and D-glucose molecules through the same MM atom, effectively "stapling" the QM moieties together. These interactions, highlighted in blue, could be misinterpreted as significant interactions in the EPD analysis, but might potentially hinder the conversion from the transition state to the product. To further elucidate these interactions, additional calculations involving pair radial distribution functions (RDFs) and noncovalent interactions (NCIs) were performed.

Solute−Solvent Behavior. The RDFs, generally denoted as $g(r)$, provide information about the probability of finding a particle at a specific distance r from another particle.[111] Hydrogen bonds (HBs) are typically identified as strong when larger $g(r)$ values occur at short distances r.[14,112] In this study, interactions were chosen to pair atoms capable of forming HBs. For instance, the O6 oxygen atom from D-glucose (i.e., acceptor atom) was paired with the hydrogen atoms from acetic anhydride (H_{C7} and H_{C10}). In the case of interactions with the D-glucose molecule, the OC, OD, and HS atom types were defined. The OC atom label corresponds to the oxygen atom within the six-membered ring, commonly identified as O5. The OD atom label represents the oxygen atoms at the hydroxyl groups, excluding O6. Lastly, the HS atom label refers to the hydrogen atoms at the hydroxyl groups, excluding HO6. To enhance the detection of significant HBs, Sankey diagrams were also computed and presented in Figure 7. The RDFs between the QM and MM regions shown in Figures S8−S19, along with the Sankey diagram, at reactant and transition states, were calculated using the TRAVIS software.[113]

Noncovalent interactions (NCI)[114−116] were evaluated for the structures of the reactant and transition states depicted in Figure 5. The electron wave functions were computed at the M06/6-311++G(d,p) level of theory using Gaussian16 software and subsequently used for NCI analysis with the NCIplot program.[115] Plots of the reduced gradient density (RGD) versus $\mathrm{Sign}(\lambda_2)\rho$ and the corresponding isosurfaces are shown in Figures S20 and S21 for reactants and in Figures S22 and S23 for transition states in ChCl/Etg and ChCl/U DESs.

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The RDFs between hydrogen atoms from the acetic anhydride (H_{C7} and H_{C10}) and solvent atoms are displayed in Figures S8 and S9. Despite their aliphatic nature, the H_{C7} and H_{C10} hydrogen atoms exhibit HBs, as suggested by the Sankey diagram shown in Figure 7 (yellow bands) for both the reactant and transition states in both solvents.

In the ChCl/Etg solvent, relevant interactions involving H_{C7}−OY (located at 2.95 Å) and H_{C7}−Cl (located at 2.87 Å) were observed in the reactant state, consistent with the NCI labeled as **H** in Figure S20. At the transition state, the $g(r)$ value for the H_{C7}−Cl is significant, while the remaining interactions do not exhibit defined peaks as shown in Figure S8c. However, their presence is supported by the NCI **T** label in Figure S22 and the Sankey diagram (Figure 7c).

Conversely, in ChCl/U DES, the dominant interaction in the reactant state is H_{C7}−Cl. Other interactions, notably, H_{C7}−OY and H_{C7}−O_{HBD}, exhibit minor peaks at distances beyond 2.8 Å. These weaker interactions are denoted by the NCI **P** and **R** labels in Figure S21. At the transition state, the H_{C7}−Cl interactions remain prominent (labeled **V** in Figure S23), and the H_{C7}−N_{HBD} interaction (labeled **T** in Figure S23) demonstrates a defined structure with a $g(r)$ value of ∼1.02 at 3.09 Å.

Significantly, the more pronounced peaks correspond to interactions between Cl and N_{HBD} atoms with H_{C7} hydrogen atoms in the ChCl/U DES (coordination number, N_{coord}, valuesof 0.44 and 1.47, respectively), suggesting established solvation dynamics. This effect is corroborated by the extended hydrogen bond lifetimes (τ_{lt}, detailed in the Supporting Information) compared to those in the ChCl/Etg DES (Tables S7 and S8). This enhanced stability likely facilitates the reaction pathway in the ChCl/U DES, contributing to the relatively lower activation barrier reported in Table 1.

Interactions involving the hydrogen atoms on the C10 carbon atom (H_{C10}) are depicted in Figure S9. Similar to observations in Figure S8, the main interactions with H_{C10} hydrogen atoms are attributed to the chloride anion; however, they show relatively low τ_{lt} as shown in Tables S5−S8. In ChCl/Etg DES, the interaction distances between H_{C10}−Cl atoms are 2.79 Å in the reactant state and 2.83 Å in the transition state. The latter interaction is weak, indicated by the NCI **X** label in Figure S22 and the τ_{lt} ∼ 2 ps. Additionally, at the transition state, a small peak but relevant interplay, labeled as **J** in Figure S22, is observed at 2.87 Å for the H_{C10}−OH interaction.

In comparison with ChCl/Etg DES, the interactions in both the reactant and transition state in ChCl/U DES show larger $g(r)$ values, providing longer stability as indicated by the τ_{lt} values. Particularly, the H_{C10}−Cl interaction located at 2.78 Å in the transition state shows the highest $g(r)$, but it also displays a weak nature, as indicated by the NCI **Q** and **Y** labels in Figure S23.

Despite the potential contribution of the H_{C10}−Cl interaction to the stability of the leaving group at the transition state in ChCl/U DES, a "staple" effect is observed for the same MM atom, engaging in a strong interaction with the HS atom from D-glucose. This effect is also depicted in Figure 5e. Throughout this work, the "staple" effect refers to a QM atom interacting with an MM atom, which in turn interacts with another atom within the QM region, thereby binding the atoms of the QM region through the atoms of the MM region. This effect, which correlates the H_{C10}−Cl interplay (NCI **Q** label in Figure S23) with the stronger HS−Cl interaction (NCI **G** label in Figure S23 and RDF in Figure S19d), can potentially hinder the decoupling of the leaving group from the transition state to the bulk, thereby generating an extra barrier for product formation. This effect may arise not only from the large N_{coord} of the chloride anion around the H_{C10} and the HS hydrogen atoms (0.44 and 0.75 in the ChCl/U DES, respectively) but also from the virtually constant presence of the HS−Cl interaction throughout the entire simulation (Table S8).

Considering that the "staple" effect emerges from two correlated distances, combined distribution functions (CDFs) are employed for further analysis. As this effect pertains to individual atoms rather than an ensemble, the CDF was computed for each chloride atom in the simulation box of the transition-state-ChCl/U DES system. The resulting CDF for H_{C10}−Cl and HS−Cl distances is depicted in Figure 8. Other pertinent CDFs are also displayed in Figures S24 and S25.

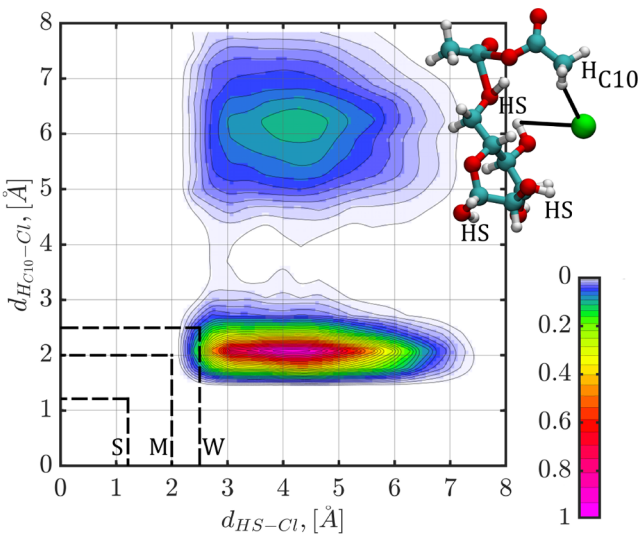


Figure 8. CDF between H_{C10}−Cl−HS atoms at the transition state in the ChCl/U DES. The combination of the H_{C10}−Cl and HS−Cl RDFs through the same Cl atom reveals the region and occurrence of the "staple" effect. Dashed boxes indicate the strength (strong, moderate, or weak) of the HB based on distance criteria.[117] Occurrence has been normalized.

Figure 8 shows a significant correlation, indicated by the red region observed between 2.8 and 5.5 Å in the HS−Cl distance (d_{HS-Cl}) and around 2 Å in the $d_{H_{C10}-Cl}$, which describes the "staple" effect. Interestingly, the "staple" effect can also be inferred from the Sankey diagram in Figure 7d, through the yellow-white band and, subsequently, by the white-orange band.

The "staple" effect is also observed in ChCl/Etg DES through the H_{C10} hydrogen atoms. However, in this case, the relevant interactions do not involve the Cl atom (as shown in Figure S24a). Instead, it is the OH oxygen atom from the HBD that plays a role in the "staple" effect through H_{C10}−OH−HS and H_{C10}−OH−H_{C7} interactions (Figures 9 and S24b). It is worth noting that the $H_{C7,C10}$−OH interactions either lack a defined structure (Figure S8c) or have lower occurrence (Figure S9c), along with shorter τ_{lt} (<0.18 ps) compared to the H_{C10}−Cl interaction in ChCl/U DES (with τ_{lt} of 3.76 ps). Furthermore, as indicated by the NCI **J** and **V** labels in Figure S22, the "staple" effect from H_{C10}−OH−H_{C7} is expected to be

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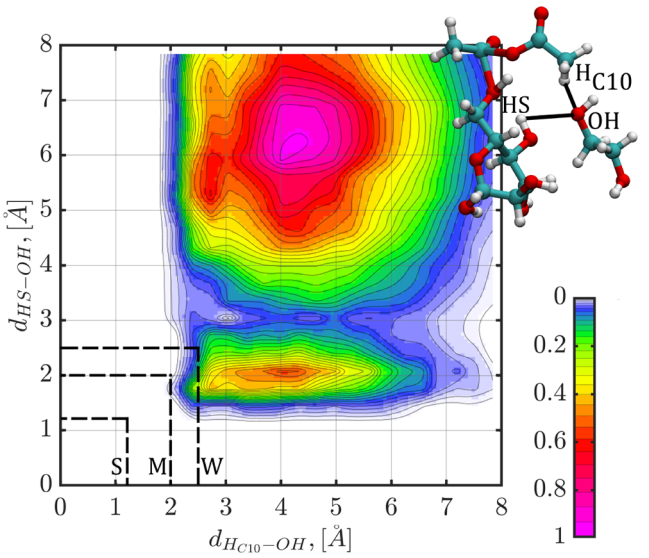


Figure 9. CDF between H_{C10}−OH−HS atoms at the transition state in the ChCl/Etg DES. Same description as in Figure 8.

weak and short. Conversely, since the H_{C10}−OH−HS "staple" effect is not particularly depicted in Figure S22, charges from the QM region are analyzed and displayed in Figures S26−S28 to provide further insights (a detailed description is available in the Supporting Information). The increasing negative charge on the C10 carbon atom during the formation of the transition state in the ChCl/Etg DES (Figure S26) is associated with the enhanced positive charge on the H_{C10} hydrogen atoms (H10, H11, and H12 in Figure S27) compared to those in the ChCl/U DES. However, given the relatively low negative charge on the OH oxygen atom (−0.5600e) compared to the chloride anion and the slightly larger H_{C10}−OH−H_{C7} distances involved in the "staple" effect, as shown in Figure S24b compared to those in Figure 8, this effect is expected to be more pronounced and longer in the ChCl/U DES.

Considering the increased negative charge on the C7 and C10 carbon atoms along the reaction path (Figures S27a and S26f, respectively), and their ability to form HBs, as observed from the Sankey diagram (light blue bands in Figure 7), their interactions with the solvent were evaluated. From the RDFs displayed in Figures S10 and S11, the interactions involving the carbon atoms exhibit some defined structures but at larger distances (>3 Å), consistent with weak interactions (for example, NCI **Q** and **R** labels in Figure S20 or NCI **O** label in Figure S23). Particularly, the C7−H_{Ch} interplay in the ChCl/U DES (3.22 Å and $g(r) \sim 1.09$) presents a longer τ_{lt} around 65 ps, probably due to the large N_{coord} of 2.36, which could provide substantial stability to the reactant state. This observation is in agreement with a large activation free energy barrier. Despite the C10−HY interaction presenting a small shoulder located at 3.09 Å, the most relevant interplay for the reactant in the ChCl/Etg DES is the C10−H_{HBD} interaction (NCI **Q** label in Figure S20), which presents a τ_{lt} value of ∼23 ps (N_{coord} of 1.47). Interestingly, this value is one-third of the τ_{lt} for the C7−H_{Ch} interaction in the ChCl/U DES, probably related to the lower N_{coord}. At the transition state, RDFs involving C10−H_{Ch} and C10−H_{HBD} interactions exhibit well-defined structures with interaction distances of around 3.5 Å in both solvents. A closer examination of Figure 5b reveals that the choline cation acts as a bridge, connecting both the C10−H_{Ch} and O7−H_{Ch} interactions, which is consistent with the "staple" effect. This effect was further corroborated through the corresponding CDF shown in Figure S24c. However, the relative neutral charge on the H_{Ch} hydrogen atoms (0.0330e), resulting in a nonlocalized C10−H_{Ch} NCI (labeled as **W** in Figure S22), may contribute to reducing the O7−H_{Ch}−C10 "staple" effect. Interestingly, the "staple" effect observed from these interactions suggests that this phenomenon can also originate from a single MM molecule. On the other hand, the absence of the "staple" effect involving C7 and C10 carbon atoms (Figure S25a,b) in ChCl/U DES suggests that their corresponding interactions do not hinder the displacement of the leaving group, providing stability to the transition state, particularly the C10−H_{Ch} interaction ($\tau_{lt} \sim 100$ ps).

As indicated by the Sankey diagram, the number of HBs involving the O7 and O9 oxygen atoms (highlighted by the dark green band in Figure 7) is up to twice the HBs observed for the C9 and C10 carbon atoms. As shown in Figure S12, the interactions involving the O7 oxygen atom are more prominent in ChCl/U DES compared to the ChCl/Etg DES at both the reactant and transition states. Interestingly, the defined peaks at the reactant state, including the O7−H_{Ch} and O7−HY interactions (NCI **S** and **N** labels in Figure S21, respectively), which yield N_{coord} values around 3.3, provide stability, suggesting a significant barrier to the formation of the transition state. This effect is particularly evident in the ChCl/Etg DES, where the O7−H_{Ch} interplay is notably reduced at the transition state, despite the O7 oxygen atom acquiring a more negative charge (Figure S26b). However, this reduction can be advantageous to the formation of the transition state since it diminishes the "staple" effect of which it conforms (Figure S24c), allowing an increase in O7−H_{HBD} interactions. The latter provides significant support to the transition state due to its longer τ_{lt} value (∼20 ps). Conversely, in the case of the ChCl/U DES, there is a substantial increase in the O7−HY interaction from the reactant to the transition state, as shown in Figure S12d (∼2.91 at 2.12 Å), providing longer support ($\tau_{lt} \sim 28$ ps) at this state. On the other hand, the O7−H_{Ch} interaction forms part of the "staple effect" in conjunction with both OC−H_{Ch} and HW−OC interplays, as shown in Figure S25e,f. Interestingly, the OC−H_{Ch} interaction shows no well-defined peaks as indicated in its corresponding RDF (Figure S17d), which further presents $\tau_{lt} < 0.16$ ps. Despite the OC oxygen atom presenting a substantial negative charge (Figure S27n), the relative neutral charge on the H_{Ch} hydrogen atoms, in conjunction with the aforementioned behavior, suggests that the OC−H_{Ch}−O7 interaction does not produce a substantial "staple" effect.

The interactions involving the O8 oxygen atom are shown in Figure S13, where this oxygen atom is part of the C8−O8 covalent bond within acetic anhydride. According to Fu et al.,[118] substantial interactions between the solute and solvent can perturb solute covalent bonds, potentially influencing their strength and facilitating dissolution within the solvent. Therefore, perturbation of this bond is a pivotal event in the reaction, contributing to the formation of the products. In the reactant state, the interactions associated with the O8 oxygen atom in the ChCl/Etg DES lack a distinct structural pattern. However, they are still present through near-to-zero NCI (**S** label in Figure S20) throughout the entire simulation ($\tau_{lt} \sim$ 440 ps). At the transition state, the previously weak O8−H_{Ch} interaction gains relevance in the ChCl/Etg DES ($\tau_{lt} \sim 115$ ps), likely aiding in the decoupling of the leaving group.

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Additional interactions are evident in ChCl/Etg DES involving HW hydrogen atoms, as depicted in Figure 5b (labeled as *b* and *c*) and further characterized by the NCI **L** label in Figure S22. Despite the fact that O8−HW contributed to the "staple" effect (Figure S24d) alongside the strong OD−H_{Ch} interaction (Figure S16c), the small atomic charge on the HW (0.0340*e*) results in a weak effect. This observation, coupled with the extremely low HBs related to the O8 oxygen atom (despite of the relevant N_{coord} of 1.29) as shown in the Sankey diagram (light brown band in Figure 7c), allows a significant concentration of electron density on such an atom. Consequently, this results in a more negative charge on the O8 oxygen atom (−0.6897*e*, Figure S26c), enhancing the attraction toward the HO6 hydrogen atom (0.4154*e*, Figure S26h), in accordance with the shorter O8−HO6 distance depicted in Figure 3 for the transition state compared to the ChCl/U DES. In the ChCl/U DES, the O8−H_{HBD} interaction (NCI **U** label in Figure S23) is the most relevant interaction toward the O8 oxygen atom (τ_{lt} ∼ 12 ps). Given the larger charge on H_{HBD}, it is proposed that the reactivity of the DES, and consequently the lower activation barriers, is enhanced when the choline cation directly interacts with the O8 oxygen atom through weak and longer HBs.

Interactions involving the O9 oxygen atom, which is crucial as part of the leaving group, are illustrated in Figure S14. In the ChCl/Etg DES, the O9−H_{Ch} interaction, which is weak but longer in the reactant state (NCI **T** label in Figure S20 and τ_{lt} ∼ 18 ps), changes from 2.61 ($g(r)$ = 1.58) to 2.75 Å ($g(r)$ = 1.33) at the transition state (NCI **U** and **S** labels in Figure S22). This weakening likely correlates with the emergence of the O9−HO interplay positioned at 1.89 Å (N_{coord} = 0.28). Conversely, in the ChCl/U DES, interactions involving the O9 oxygen atom at the transition state (NCI **R** and **J** labels in Figure S21) are lower compared to those found at the reactant state. The lack of stronger support for the O9 oxygen atom likely results in a series of intermolecular contacts within the transition state, which can hinder product formation. These contacts are labeled as **B**, **N**, and **K** in Figure S22 for ChCl/Etg DES and as **B**, **F**, and **K** in Figure S23 for ChCl/U DESs. Furthermore, in Figure S24e, the O9 oxygen atom participates in the "staple" effect in the ChCl/Etg DES (NCI **L**, **U**, and **Y** labels in Figure S22), particularly through the longer O9−H_{Ch} interaction. Interestingly, the O9−HO interplay assists in compensating for this effect, as observed from the Sankey diagram. However, this is not true for the same state in ChCl/U DES, as shown in Figure 5e, where the "staple" effect is formed through longer O9−H_{HBD} (τ_{lt} ∼ 290 ps) and O9−H_{Ch} (τ_{lt} ∼ 22 ps) interplays in Figure S25c,d. These interactions were further confirmed through NCI calculations (**R** and **J** labels in Figure S23). Consequently, the large number of interactions favoring the "staple" effect in the ChCl/U DES is likely one of the reasons behind the increased free activation barrier compared to the ChCl/Etg DES.

Another pivotal interaction involves the O6 oxygen atom, as depicted in Figure S15 and indicated by the turquoise band in the Sankey diagram. The RDFs for the O6 oxygen atom illustrate substantial interactions, primarily with choline-derived hydrogen atoms (H_{Ch} and HY), with a τ_{lt} around 15 ps at the reactant state. In the ChCl/Etg DES in the reactant state, the O6−H_{Ch} interaction emerges as the most significant interaction among others. This interaction is characterized as weak (NCI **U** label in Figure S20) due to its considerable interaction distance (3.09 Å) and the small atomic charge on the H_{Ch} hydrogen atom (0.0330*e*), regardless of the substantial negative charge on the oxygen atom (Figure S26g). At the transition state in ChCl/Etg DES, RDFs reveal no well-defined local structures for the O6 interactions; however, as noticed in Figure 7b, there is a slightly large HB count. Considering that the acetylation reaction is a nucleophilic reaction and a large positive charge on the C8 carbon atom (Figure S26a, ∼0.7854*e*), the low number of O6−H interactions could obstruct the nucleophilic attack of the O6 oxygen atom from the reactant to the transition state. Therefore, weak but competing interactions, for instance, the O6−H_{Ch} that shows N_{coord} = 9.14 and τ_{lt} around 115 ps, could increase the activation barrier. In the reactant state within the ChCl/U DES, both the O6−H_{Ch} and O6−HY interactions assume crucial roles. The presence of these interactions, particularly O6−HY at 2.16 Å indicates potential steric hindrance (N_{coord} = 1.12), impacting the nucleophilic attack of the O6 oxygen atom on the C8 carbon atom in ChCl/U DES. However, at the transition state, this interaction is located at 4.25 Å (N_{coord} = 0.55). Although the more positive charge (Figure S26g) weakens the nucleophilic attack toward the C8 carbon atom at the transition state, the steric effects are also significantly reduced (as observed in Figure 7d), presenting shorter τ_{lt} in comparison to the ChCl/Etg DES. This effect allows the formation of the transition state in ChCl/U DES, resulting in a relatively low activation barrier.

Interactions involving the HO6 hydrogen atom are illustrated in Figures S16 and 7 (green band). Interestingly, as observed from Figure 7, the HO6 hydrogen atom forms a number of HBs in the reactant state; however, they are drastically reduced in the transition state. In the ChCl/Etg DES system (Figure S16a), the HO6 hydrogen atom predominantly engages with the Cl atom at the reactant state through strong interactions (NCI **B** label in Figure S20, N_{coord} = 0.33). Conversely, in the ChCl/U DES system, it interacts with OY oxygen atom with N_{coord} of 1.62. At the transition state, the most prominent NCI (**A** label in Figures S21 and S23) involves the HO6 hydrogen atom. In the case of ChCl/Etg DES, the HO6−O4 interaction likely facilitates the decoupling of HO6 from the O6 oxygen atom toward the O8 atom. Additionally, the HO6 hydrogen atom predominantly engages with the OY oxygen atom from choline cation at 3.49 Å. The increased distance and the short τ_{lt} (<0.1 ps) for this interaction facilitate unimpeded access for the HO6 hydrogen atom, resulting in a more pronounced interaction with the O8 oxygen atom. Conversely, in the ChCl/U DES, the decoupling of the HO6 is also supported by the O4 oxygen atom. However, this interaction is of a weak nature (see the NCI **H** label in Figure S23). As observed from Figure S16d, the HO6 hydrogen atom engages mainly with the Cl atom (2.27 Å, N_{coord} = 0.98). Despite this interaction contributing to the decoupling of the HO6 hydrogen atom, there is an increased distance and weakened O8−HO6 interaction at the transition state in the ChCl/U DES, as depicted in Figure 4. This effect is probably caused by the steric effect raised from the chloride anion and even by the N_{HBD} nitrogen atom (located at 3.54 Å), which show longer τ_{lt} (221 and 40 ps, respectively).

Finally, the RDFs for the OD, OC, and HS atoms from D-glucose (Figures S17 and S18) presented well-defined peaks and elongated and stronger interactions in both solvents. Particularly, the most prominent HS−Cl interactions are also depicted in the NCI surfaces (**A**−**E** labels in Figures S20 and

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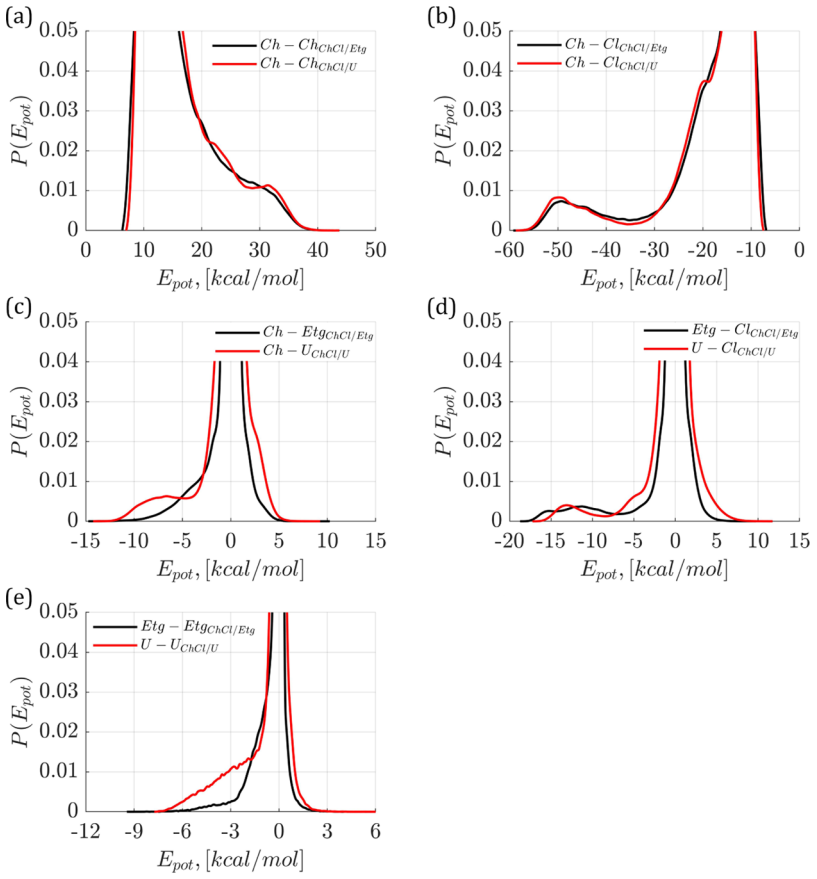


Figure 10. Energy pair distribution (EPD) between (a) Ch−Ch, (b) Ch−Cl, (c) Ch−HBD, (d) HBD−Cl, and (e) HBD−HBD closer molecules to the reactant state within the ChCl/Etg (black line) and the ChCl/U (red line) DESs.

S21), which are present along the trajectory. This further aligns with the large white-orange band in Figure 7a,b. The RDFs regarding the OC and OD oxygen atoms reveal that solvation is enhanced at the transition state. However, as indicated in Figures S24 and S25, both atoms are part of the "staple" effect. A detailed discussion of these interactions is available in the Supporting Information.

Notably, these nuanced observations not only deepen our understanding of the intricate molecular dynamics but also underscore the validity and efficacy of the chosen reaction coordinate in computing free energy barriers. To further unravel the role of the solvent, additional properties, including energy pair distribution, diffusion coefficients, HB lifetimes, and correlations with experimental Kamlet−Taft parameters, were also evaluated.

Role of the Solvent. Xia et al.[119] and Zhang et al.[120] concluded from their respective studies that the strength of interactions within the DES significantly impacts the solute−solvent interactions. Therefore, the relationship between the interaction strength within the evaluated DES and the reported free energy barriers in Table 1 was also investigated.

Upon introduction of the solute into the solvent, a series of changes in the molecular arrangement of the solvent close to the QM region become evident, as observed from Figure S29. These changes are associated with the energetic interactions between DES molecules in the reactant-DES systems. Accordingly, energy pair distributions (EPDs) were computed for the interactions between choline cations (Ch−Ch, Figure 10a), choline and chloride ions (Ch−Cl, Figure 10b), choline and HBD molecules (Ch−HBD, Figure 10c), HBD and chloride molecules (HBD−Cl, Figure 10d), and HBD molecules (HBD−HBD, Figure 10e) around the QM region within both ChCl/Etg and ChCl/U DESs. To address the variable number of molecules around the QM region in both solvents, the distributions were normalized, yielding probability values ranging from 0 to 1.

Figure 10a,b illustrates that both the Ch−Ch and Ch−Cl EPDs exhibit similar behavior in both solvents, with minor fluctuations. This effect is a consequence of their similar RDFs around the QM region, as displayed in Figure S29a,d, which are similar to those obtained in the pure-DES system (see details in the Supporting Information). The difference between the pure-DES and reactant-DES $g(r)$ values emerges because in the reactant-DES system, the solvent is diluted due to the presence of the solute in comparison with the pure solvent.[121] Further analysis indicates that a similar behavior in Figure 10a,b may arise from the choline cation conformation, which solely displays the *trans* conformation related to the NA−OY dihedral, as shown in Figure S30a. Interestingly, the OY−HY interaction is similar between both mixtures (Figure S29b), but they differ from the pure state, reflecting the distinct roles that both atoms play around the QM region.

The Ch−U strong interactions are significantly more prevalent compared to Ch−Etg EPD, as shown in Figure 10c. This phenomenon stems from the fact that, at short range, the pair choline cation and urea is closer than the pair choline−ethylene glycol, as depicted in Figure S29c. This behavior does not entail changes in the conformational structure of the HBD molecules, as shown in Figure S30b,c from the pure state. Interestingly, the Ch−U RDF shows a

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second peak around 8.6 Å, which supports its bulk structure. Stefanovic et al.[122] previously reported similar enhanced interactions between the choline cation and urea HBD, relating such interactions to a greater melting point depression and higher shear viscosity of the ChCl/U DES compared to other ChCl-based solvents, including the ChCl/Etg DES. The reinforced interactions between the choline cation and urea HBD impose a mutual constraint for both molecules in establishing new HBs with the solute. This reduces the probability of the "staple" effect emerging, unlike their counterparts in the ChCl/Etg DES. In this regard, as observed in Tables S3 and S4, the τ_{lt} of the Ch−Etg ranges up to 31 ps, while that of the Ch−U reach up to 126 ps. The increased τ_{lt} in the ChCl/U DES enhances the probability to find strong interaction energies, consistent with the observations in Figure 10c. Conversely, interactions involving HBs between choline and ethylene glycol can readily adapt to changes in the solute, enhancing solute−solvent interactions.

From Figure 10d, the EPD for the HBD−Cl exhibits slightly stronger interactions within the ChCl/Etg DES compared to the ChCl/U DES (see also RDF in Figure S29f), characterized by two regions at −11.2 and −15.4 kcal/mol. This behavior particularly emerges from the HO−Cl interactions (positioned at 2.07 Å, Figure S29g), which present a τ_{lt} value of around 50 ps. In contrast, the U−Cl interplay exhibits a relevant peak located at −13 kcal/mol. The relatively low interacting energy in the U−Cl EPD compared to the Etg−Cl EPD likely enables reliable interactions with the solute (NCI **A**, **C**, **D**, **F**−**H**, **K** labels in the Figure S21), despite the large τ_{lt} (∼180 ps). This is possible because the U−Cl interactions are associated with the second solvation shell around 5.15 Å in Figure S29h (N_{coord} ∼ 1.27). Therefore, allowing the remaining H_{HBD} acidic hydrogen atoms to adopt an in-plane conformation (Figure S30c) to follow the chloride anion, maintaining the HB network through weaker interactions,[122] but still available to create other interactions. This effect is less favored in the ChCl/Etg DES, which possesses only two acidic hydrogen atoms (HO) that can be found in either the *gauche* or *trans* conformations (Figure S30b). Consequently, it limits HBs through HO hydrogen atoms despite the shorter τ_{lt}, as evidenced by the pink bands in Figure 7a,b. Conversely, since the U−Cl interactions are relatively weaker, the chloride anion might have greater freedom to interact with the QM region, producing the "staple" effect. In this regard, the Cl−Cl RDF in ChCl/U DES (Figure S29j) suggests a reorganization of chloride anions due to the presence of a second peak located around 7.5 Å, increasing the coordination number and then the probability that the "staple" effect is present in such a system. On the other hand, the RDF between chloride anions in the ChCl/Etg DES suggests an increase in their repulsions. This is evident from the short distance position of the peak in the reactant-DES system with no changes in its RDF shape compared with the pure state. This effect may support the minor role of the anion in the "staple" effect.

The HBD−HBD EPD, shown in Figure 10e, indicates that the interactions between urea units (U−U) are more pronounced than those between ethylene glycol units (Etg−Etg), as is also displayed in Figure 7a,b. This behavior predominantly stems from the defined HBD−HBD RDF shown in Figure S29e and by the significant H_{HBD}−O_{HBD} interaction in Figure S29i, which shows a longer τ_{lt} (∼70 ps). Notably, this interaction exhibits a coordination number of 1.58. Conversely, although the Etg−Etg interactions show higher interaction energies (reaching up to −9.5 kcal/mol), these interactions are less probable due to the low N_{coord} calculated for the reliable HO−OH interaction (Figure S29i), which stands at 0.14. Consequently, the urea HBD exhibits a diminished capacity to form HBs with the solute through the O_{HBD} oxygen atom.

The significant difference in the average τ_{lt} between ChCl/U DES (68.30 ps) and ChCl/Etg (14.01 ps), along with the higher energies presented in the EPD, aligns with the experimental viscosity values extrapolated at 373.15 K (17.06 and 5.87 cP) for ChCl/U and ChCl/Etg DES, respectively.[123,124] This correlation suggests that as the viscosity increases, the efficiency of solute−solvent interactions diminishes, introducing additional energy barriers for processes such as solubility[125] and chemical reactions.[32] Considering that wood acetylation primarily depends on the diffusion of acetic anhydride,[14] an assessment of the diffusion coefficient (D) of the involved species in ChCl/Etg and ChCl/U DESs was conducted. For this purpose, additional NVT simulations for pure-DES and acetic-anhydride-DES lasting 10 ns at 373.15 K, employing the same MM and QM/MM levels of theory in this study, were conducted for both solvents. In the case of the acetic-anhydride-DES systems, a single molecule of solute was employed. The self-diffusion coefficients of choline, chloride, HBDs, and acetic anhydride were computed using the mean square displacement (MSD) with the VMD Diffusion Coefficient Tool plugin.[126]

Before calculating the diffusion coefficient, the MSD was used to determine the time-dependent β-parameter ($\beta(t)$).[127] This step accurately identifies the diffusive regime, facilitating their determination.[40] Figures S31 and S32 depict the variation of the $\beta(t)$ parameter value over time for the involved species. The appropriate intervals for calculating the diffusion coefficients, $\beta(t) \sim 1$, are displayed in Table S9. Computed diffusivities were adjusted for system-size effects[128] using the experimental shear viscosity extrapolated from refs 123,124. The calculated diffusion coefficient for acetic anhydride was 1.43×10^{-10} m^2/s in ChCl/Etg DES and 0.69×10^{-10} m^2/s in ChCl/U DES. The larger diffusion coefficient observed for acetic anhydride in ChCl/Etg DES aligns with the lower activation barrier reported in Table 1, suggesting that the barrier imposed by the HB network within that solvent is less pronounced compared to that in ChCl/U DES, in accordance with the abovementioned discussion.

From Table S9, it is evident that the chloride anion and HBDs present higher diffusion coefficients compared with the choline cation in both solvents. This suggests that interactions between the choline cation and the acetate moiety (C7, C8, O7, and H_{C7}) or D-glucose can act as anchors. This effect is evident for the reactant in ChCl/U DES in Figure S21 (NCI **N**, **P**, **R**, and **S** labels). At the transition state, the increased O7−HY interaction also supports this state by anchoring the acetate moiety. This is a consequence of the more negative charge on the O7 oxygen atom , as discussed in the QM charge analysis section (Supporting Information). Furthermore, the OC−H_{Ch}−O7 "staple" effect that emerged in the ChCl/U DES does not represent an impediment to the conversion, as it is acting as an anchor for the acetylated glucose. This effect can explain the larger reactant solvation and the relatively low activation barrier in the ChCl/U DES. In contrast, the anchor effect is less supported for the reactant in the ChCl/Etg DES, as observed in Figure S20 (NCI **H**, **N**, and **S** labels). At the transition state, the anchor effect through the choline cation is

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evident for the same interactions that describe the "staple" effect shown in Figure S25e,f. However, as the OC−H_{Ch} interaction shows no defined structure, anchoring could be reduced.

Given that the acetate moiety is better supported in the ChCl/U DES and considering that the $\Delta G^{\ddagger}$ is lower in the ChCl/Etg DES, it is anticipated that the acetic acid moiety (C9, C10, O8, O9, and H_{C10}) or leaving group exhibits larger solvation in ChCl/Etg DES. In this regard, interactions pertaining to the acetic acid moiety through the faster chloride anion and HBD can act as pullers along the reaction coordinate, mainly at the transition state. This can be observed in Figure S22, where the leaving group is supported by the substantial O9−HO interplay (NCI **S** label in Figure S22), along with noncovalent interactions **J**, **P**, and **X** in the same figure, despite the "staple" effect being present at the **J**, **L**, and **U** interactions. In the case of the ChCl/U DES, the leaving group is supported by the **Q**, **R**, **U**, and **Y** puller interactions displayed in Figure S23. However, the **Q** interaction is also involved in the strong "staple" effect (Figure 8).

Relation with Kamlet−Taft Parameters. In the preceding discussion, it has been emphasized that a comprehensive understanding of the behavior of HBs between molecules within the DES is fundamental to fully comprehend the effectiveness of the solvent in chemical reactions. Kakuchi et al.[90] suggested that the basicity of ILs might enhance transesterification reactions. In this context, the Kamlet−Taft (KT) parameters, which describe solvent polarity, polarizability, acidity, and basicity, offer valuable insights into hydrogen bonding capabilities. The KT "α" parameter quantifies hydrogen bond acidity (ability to donate protons), while the KT "β" parameter quantifies hydrogen bond basicity (ability to accept protons).[129] Despite their importance, KT values for DES are scarce.[130] Early compendia of experimental KT parameters for structurally diverse DES were reported by Florindo et al.,[131] and more recently by Zhang et al., who correlated them with the capacity to dissolve lignocellulosic components.[132] The KT α and β parameter values for ChCl/Etg and ChCl/U DESs from both investigations are summarized in Table 2.

Table 2. Experimental Reported Kamlet−Taft (KT) α and β Parameters for the Evaluated DES

solvent	α	β	$\beta-\alpha$[a]
ChCl/Etg	1.47[131]	0.57[131]	−0.900
	0.741[132]	0.555[132]	−0.186
ChCl/U	1.42[131]	0.50[131]	−0.920
	0.992[132]	0.477[132]	−0.515

[a]"Net basicity", as defined by Zhang et al.[132]

As observed, the KT β values exhibit similarity between reference studies, showing higher values for the ChCl/Etg DES and lower values for the ChCl/U DES. Conversely, a notable distinction is evident for the KT α parameter. Particularly, despite these differences, it is essential to highlight that a higher KT β parameter correlates with a lower activation free energy, as detailed in Table 1. In contrast, no direct correlation is observed between the $\Delta G^{\ddagger}$ and KT α parameter from both references. Given the intricate relationships associated with HBs, as elucidated by the EPD between DES molecules, it is anticipated that a more intricate descriptor could emerge from their KT parameters. Surprisingly, upon evaluating the net basicity ($\beta-\alpha$), as defined by Zhang et al.,[132] a direct correlation with $\Delta G^{\ddagger}$ is identified for both sets of KT parameter values. In this context, a noteworthy trend emerges: the greater the net basicity ($\beta-\alpha$), the lower the $\Delta G^{\ddagger}$. This observation suggests that reduced activation barriers for the present reaction could potentially be achieved by having larger and more similar KT α and β parameters (i.e., $\beta-\alpha$ approaching zero) or even when $\beta-\alpha$ becomes positive. This finding is further consistent with the presence of both a larger HB network and "staple" effects in the ChCl/U DES, in addition to the lower diffusion coefficient for the acetic anhydride. Therefore, this outcome underscores the potential for selecting or designing efficient DES solvents through an initial screening based on their KT parameters and their ability to form HBs. However, it is crucial to emphasize that experimental validation is necessary to confirm or challenge this hypothesis.

Formation of Choline-Chloride Acetic Acid DES. The acetylation reaction of LCB generates acetic acid as a byproduct[1,5,11] (Figure S3b), which can catalyze the hydrolysis of biomass.[26] Accordingly, the timely formation of a choline-chloride acetic acid (ChCl/AA) DES mixture upon the production of acetic acid could potentially inhibit both cellulose and lignin depolymerization as well as corrosion in equipment and pipes. For this purpose, the interaction energy (ΔE_{int}) of the DES cluster was analyzed to gain insights into the formation and stability of ChCl/AA at 1:1 and 1:2 molar ratio DESs in comparison to the ChCl/Etg and ChCl/U DESs. The interaction energy was evaluated using the following expression[133]

$$\Delta E_{int} = E_{DES\ cluster} - (E_{HBA} + E_{HBD}) \quad (3)$$

where $E_{DES\ cluster}$ is the potential energy of the optimized DES cluster, E_{HBA} is the potential energy of the ChCl ion pair (HBA), and E_{HBD} corresponds to the potential energy of the HBD molecules. Both E_{HBA} and E_{HBD} were computed by employing the geometry of the complex. The initial geometries for the interaction energy calculations were derived from a classical MD equilibration of individual DES clusters for both ChCl/Etg and ChCl/U. For the ChCl/AA DES, the cluster was constructed using the HBD conformation from the ChCl/Etg DES cluster. Particularly, for the ChCl/AA DES, two different molar ratios were examined: 1:1 and 1:2. Geometry optimizations and single-point energy calculations using the M06/6-311++G(d,p) level of theory were performed using the Gaussian16 package. No imaginary frequencies were observed for any of the optimized structures. The results for ΔE_{int} are presented in Table 3.

The results obtained for the ChCl/Etg and ChCl/U cluster DESs agree with those reported by Wagle et al.[133] Notably, the data in Table 3 reveal that the ΔE_{int} of the ChCl/AA at 1:2 molar ratio is comparable to that of ChCl/Etg and ChCl/U

Table 3. Computed Interaction Energies (ΔE_{int}, in kcal/mol) for ChCl-Based DES Clusters[a]

cluster (HBA/HBD molar ratio)	ΔE_{int}
ChCl/Etg (1:2)	−41.42
ChCl/U (1:2)	−42.59
ChCl/AA (1:1)	−15.29
ChCl/AA (1:2)	−42.08

[a]No additional energy corrections were used.

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DESs. This suggests that acetic acid could interact with the choline-chloride HBA, leading to the formation of the ChCl/AA (1:2) DES mixture, which exhibits a more negative interaction energy compared to the ChCl/AA (1:1) DES. These observations imply that the formation of ChCl/AA (1:2) DES might significantly reduce the ability of acetic acid to degrade the wood structure via hydrolysis reactions and substantially mitigate equipment corrosion.

CONCLUSIONS

Hybrid quantum mechanics (QM) molecular mechanics (MM) simulations were employed to evaluate the potential of mean force for the D-glucose acetylation reaction with acetic anhydride within choline-chloride ethylene glycol (ChCl/Etg) and choline-chloride urea (ChCl/U) deep eutectic solvents (DES). The calculated activation free energy barriers were found to be 18.19 ± 0.15 kcal/mol for ChCl/Etg and 19.83 ± 0.15 kcal/mol for ChCl/U DES, which are consistent with the values reported in the existing literature.

Analysis of the results indicated that the acetylation reaction is favored in the ChCl/Etg DES due to larger interaction energy of the acetic anhydride moiety in the transition state compared to the reactant state, contrasting with that observed in ChCl/U DES. Examination of atomic charges along the reaction coordinate suggested that the increased negative charge on the O6 oxygen atom enhances the nucleophilic attack, particularly when the choline cation forms weak and extended interactions with the O8 oxygen atom. Additionally, interactions involving the O6 oxygen and HO6 hydrogen atoms suggested that steric effects might affect the nucleophilic attack on the C8 carbon atom.

The radial and combined distribution functions revealed the presence of a phenomenon termed here as the "staple" effect in both solvents, introducing additional barriers for product formation. This effect was more pronounced in ChCl/U DES, aligning with the larger activation free energy barrier. The "staple" effect typically comprises weak but longer hydrogen bonds, as revealed by noncovalent interactions and hydrogen bond lifetime calculations. By analyzing the role of the solvent, diffusion coefficient calculations revealed that interactions between the leaving group and hydrogen bond donor or chloride anion can act as pullers, enhancing the reaction. Additionally, interactions involving the acetate moiety through the choline cation can act as anchors, further enhancing the reaction. In this context, the diffusion coefficients of the DES molecules can serve as valuable parameters for screening potential new and efficient solvents for wood acetylation.

The energy pair distributions and radial distribution functions involving the closest solvent molecules around the QM region indicate that the strengthened interactions between the choline cation and urea led to mutual constraints for both molecules to establish new hydrogen bonds with the solute, thereby reducing the probability of the "staple" effect emergence. This effect is less extended in the ChCl/Etg DES. Conversely, the shorter hydrogen bond lifetimes found in the ChCl/Etg DES allow for a rapid response to the changes present in the solutes, resulting in relatively few restrictions on the formation of hydrogen bonds.

The Kamlet–Taft β parameter negatively correlates with the activation free energy barriers. Surprisingly, by evaluating the Kamlet–Taft net basicity ($\beta-\alpha$), the same relation was found. In this regard, lower activation barriers for the present reaction could potentially be achieved by solvents with larger and more similar Kamlet–Taft α and β parameters.

Finally, the calculation of the interaction energy suggested that forming ChCl/AA DES at a 1:2 molar ratio is plausible. This mixture could reduce side-effects related to acetic acid, such as the degradation of wood products and equipment corrosion, by capturing the acid as soon as it is produced.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.4c03585.

Atomic labels and molecular structures evaluated in this work and preparation of inputs, simulation details, and evaluation of DES properties in GROMACS and AMBER (Figure S1); radial distribution functions between ChCl and ethylene glycol from GROMACS and AMBER simulations (Figure S2); thermodynamic and structural results (Tables S1 and S2); energy-minimized configuration of the reactant (D-glucose and acetic anhydride) and product (acetylated D-glucose at C6 position and acetic acid) (Figure S3); simulation details for the minimum energy path along with selected MEP configurations for the QM region (Figure S4); potential energy against time plots for the QM/MM MD NVT equilibration (Figure S5); final configuration of the reactant-DES and product-DES systems (Figure S6); energy pair distribution for QM moieties and DES molecules (Figure S7); RDFs between QM atoms and MM atoms at reactant and transition states (Figures S8–S19); reduced gradient density versus Sign(λ_2)ρ and reduced gradient density isosurfaces for both the reactant and transition states and the closest solvent molecules (Figures S20–S23); combined distribution functions displaying the "staple" effect (Figures S24 and S25); QM charges calculation details and related discussion along with their corresponding behavior along the reaction coordinate and comprehensive discussion related to the role of OC, OD, and HS atoms at both reactant and transition state (Figures S26–S28); details of hydrogen bond lifetime calculations along with the results for relevant observations (Tables S3–S8); details for nonperiodic RDFs calculations between DES molecules in the pure-DES and reactant-DES calculations (Figure S29); conformational analyses of choline and hydrogen bond donors in both pure and reactant-DES systems (Figure S30); calculated time-dependent β-parameter ($\beta(t)$) for DES molecules (Figure S31); same for acetic anhydride (Figure S32); and computed self-diffusion coefficients (Table S9) (PDF)

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Notes

The authors declare no competing financial interest.

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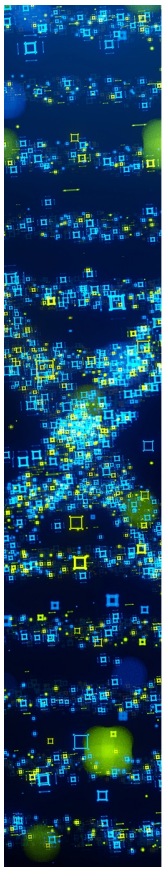
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